# EED-201 TUTORIAL-10 Answers

**Ans-①**

(i) $X(z) = -\frac{1}{6}\frac{z}{(z-1/2)(z-1/3)}$ ; $\frac{1}{3} < |z| < \frac{1}{2}$

(ii) $X(z) = \frac{a^2-1}{a}\frac{z}{(z-a)(z-1/a)}$ ; $a < |z| < \frac{1}{a}$

(iii) $X(z) = 5z^2 + 3z - 2 + 4z^{-2} - 3z^{-3}$ ; ROC: entire z except $z = 0$ & $\infty$

(iv) $X(z) = \frac{z}{(z-a)^2}$ ; $|z| > |a|$

(v) $X(z) = 2z^2 - 3z^{-2}$ ; ROC: $0 < |z| < \infty$

(vi) $X(z) = \frac{2z}{(z-a)^3}$ ; $|z| > |a|$

(vii) $X(z) = \frac{2z^{-1}}{1-2z^{-1}} + \frac{z^{-1}/4}{1-\frac{1}{4}z^{-1}}$ ; $1/4 < |z| < 2$

**Ans-②** (i) $x[n] = \{\ldots, 15, 7, 3, 1, \underset{\uparrow}{0}\}$

(ii) $x[n] = \{0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \ldots\}$

**Ans-③** (i) $x[n] = (1 - 2^n + n2^{n-1})u[n]$

(ii) $x[n] = 3(2)^{n-1}u[n-1]$

(iii) $x[n] = 2\delta[n+1] + \frac{3}{2}\delta[n] + (1-2^{n-1})u[-n-1]$

(iv) $x[n] = [(-1)^{n-1} - (-3)^{n-1}]u[n-1] + \frac{1}{2}[(-1)^{n-3} - (-3)^{n-3}]u[n-3] + \frac{3}{2}[(-1)^{n-5} - (-3)^{n-5}]u[n-5]$

Ans-4 $x[n] = (n+1)\, a^n u[n+1]$.

Ans-5 $H(z) = \dfrac{z^2}{(z-1/2)(z-1/4)}$ ; $|z| > \frac{1}{2}$

$$h[n] = \left[ 2\left(\tfrac{1}{2}\right)^n - \left(\tfrac{1}{4}\right)^n \right] u[n].$$

$$s[n] = \left[ \tfrac{8}{3} - 2\left(\tfrac{1}{2}\right)^n + \tfrac{1}{3}\left(\tfrac{1}{4}\right)^n \right] u[n]$$

Ans-6 $|k| < 2$ $\because$ ROC : $|z| > |k/2|$

For stable, ROC contains unit circle

Ans-7 $x[n] = \{ \ldots, -\frac{1}{8}, -\frac{1}{4}, -\frac{1}{2}, -1, -1, -1, \ldots \}$

Ans-8 $x[0] = 2$ ; $x[\infty] = 0$

Ans-9 (a) Difference eq$^n$ –

$$y[n] = x[n] + \frac{9}{8} x[n-1] - \frac{1}{3} y[n-1] + \frac{2}{9} y[n-2]$$

(b) Stable $\because$ ROC includes unit circle $\therefore$

ROC $\Rightarrow$ $|z| > 2/3$. {causal}

Ans-10 (i) $x[n] = \left(\frac{1}{2}\right)^2 u[n]$

(ii) $x[n] = \delta[n] - \left(-\frac{1}{2}\right)^{n-1} u[n-1]$

Ans-11 $x[n] = \frac{1}{n} a^n u[n-1]$.